Rajyashree Sen, Ph.D.

CURRICULUM VITAE

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EDUCATION

2012 – 2017 **Ph.D., Molecular Biology & Neuroscience,** *University of Vienna, Austria & Janelia Research Campus, USA*
Dissertation: Neuronal control of backward walking in *Drosophila melanogaster*
Advisor: Dr. Barry Dickson

2010 – 2012 **M.Sc., Biotechnology,** *University of Hyderabad, India*
Dissertation: Role of Hsp82 in the DNA damage response pathway
Advisor: Dr. Sunanda Bhattacharyya

2007 – 2010 **B.Sc., Microbiology,** *St. Xavier's College Kolkata, India*

PROFESSIONAL EXPERIENCE

12/2024 – present **Associate Research Scientist**, *Zuckerman Institute of Columbia University*

04/2019 – 11/2024 **Postdoctoral Research Scientist**, *Zuckerman Institute of Columbia University & Howard Hughes Medical Institute (HHMI)*
A new code for position: Generalizable spatial maps in mouse medial frontal cortex
Advisor: Dr. Richard Axel

01/2018 – 03/2019 **Postdoctoral Research Associate,** *Janelia Research Campus, USA*
Distributed control of motor circuits for backward walking in *Drosophila*
Advisor: Dr. Barry Dickson

10/2018 – 12/2018 **Invited Visiting Scientist,** *Institute of Zoology at University of Cologne*
Quantitative analysis of joint kinematics in backward walking flies
Advisors: Dr. Ansgar Büschges & Dr. Barry Dickson

05/2011 – 07/2011 **IASc-INSA-NASI Summer Fellow**, *Center for Cellular and Molecular Biology, Hyderabad (CCMB), India*
Expression and Purification of FAAL10 from *Mycobacterium tuberculosis*
Advisor: Dr. Rajan Sankarnarayanan

HONORS

Research

07/2025 – 01/2027 **Alzheimer's Disease Research Center Research Education Scholar**
Alzheimer's Disease Research Center, Taub Institute, Columbia University

01/2025 – 01/2027 **BBRF Young Investigator Grant** ($70,000)
Brain and Behavior Research Foundation

10/2012 – 09/2015 **Marie Curie Graduate Fellowship**
Marie Curie FP7 Programme FLiACT Training Network

08/2010 – 07/2012 **Fellowship for Masters in Biotechnology**
Full tuition, Department of Biotechnology (DBT), Government of India

05/2011 – 07/2011 **Summer Research Fellowship**
Indian Academy of Sciences-Indian National Science Academy-The National Academy of Sciences (IASc-INSA-NASI)

Pedagogy

03/2025, 07/2025 — **Teresa Vilardi Scholarship**
Institute of Writing and Thinking, Bard College

Public Engagement

02/2025 – 12/2026 — **The Puffin Foundation Annual Artists Grant** ($2500)
The Puffin Foundation Ltd., New Jersey

01/2025 – 07/2025 — **Research! America Civic Engagement Microgrant** ($1800)
Research! America, Arlington, Virginia

10/2023 – 05/2024 — **Outreach and Public Engagement Seed Grant** ($750)
Zuckerman Institute's Public Engagement Program, Columbia University

Project: The Lab: Bridging Science and Community through Improv Comedy. Used the above grants to conceive, found and produce improv comedy shows inspired by live interviews with neuroscientists. The goal of this project is to engage the general public in scientific discourse. The project was featured in Columbia Magazine in September 2026.

PUBLICATIONS

In Preparation

Rajyashree Sen, Lorenzo Posani, Alexander Medoff, Baihe Zhang, Larry Abbott, Richard Axel*, Stefano Fusi* (2026). "A new neural code for position: generalizable spatial maps in mouse medial frontal cortex"**.** ***bioRxiv.***
DOI: https://zp8urt.s.gy/z6mhnK **co-corresponding authors*

Preprints

Valentine Andreu, **Rajyashree Sen**, Nour El Houda Mimouni, Eun Ji Lee, Dianne-Lee Ferguson, Alexis Stutzman, Bianca J. Marlin (2025). "Early postpartum development of pup urine preference in mothers"**.** ***bioRxiv***.
DOI: https://doi.org/10.1101/2025.09.01.673527

Peer Reviewed

Kai Feng, **Rajyashree Sen**, Ryo Minegishi, Michael Dübbert, Till Bockemühl, Ansgar Büschges, Barry J Dickson (2020). "Distributed control of motor circuits for backward walking in *Drosophila*" ***Nature Communications***. 11(6166).
DOI: https://doi.org/10.1038/s41467-020-19936-x

Rajyashree Sen, Kaiyu Wang, and Barry J. Dickson (2019). "Twolumps ascending neurons mediate touch-evoked reversal of walking direction in *Drosophila*" ***Current Biology***. 29(24): 4337-4344.e5.
DOI: https://doi.org/10.1016/j.cub.2019.11.004

Rajyashree Sen, Ming Wu, Kristin Branson, Alice Robie, Gerald M. Rubin and Barry J. Dickson (2017). "Moonwalker descending neurons mediate visually evoked retreat in *Drosophila*" ***Current Biology***. 27(5): 766-771.
DOI: https://doi.org/10.1016/j.cub.2017.02.008

PRESENTATIONS

11/2026 — **Poster |** A new code for position - Euclidean neural maps in the mouse medial frontal cortex **(forthcoming)**
Rajyashree Sen, Alexander Medoff, Lorenzo Posani, Larry Abbott, Richard Axel, and Stefano Fusi.
Society for Neuroscience (SfN), Washington D.C., USA

10/2026 **Selected Talk |** A linearly generalizable spatial map in mouse medial frontal cortex
Geometry in Neuroscience Workshop, National Institute of Theory and Mathematics in Biology, USA

09/2026 **Invited Talk |** A new code for position: flat neural maps in mouse medial frontal cortex
Southern Illinois University, USA

08/2026 **Poster |** A new code for position: flat neural maps in mouse medial frontal cortex
Rajyashree Sen, Alexander Medoff, Lorenzo Posani, Larry Abbott, Richard Axel, and Stefano Fusi.
Cognitive Computational Neuroscience Conference, New York University, USA

03/2026 **Poster |** The dorso-medial prefrontal cortex abstracts space onto a low-dimensional manifold
Rajyashree Sen, Alexander Medoff, Lorenzo Posani, Larry Abbott, Richard Axel, and Stefano Fusi.
Computational and Systems Neuroscience (COSYNE) Conference, Portugal

06/2024 **Nominated Talk |** Combined coding of individual identity and spatial locations in the mouse dorso-medial prefrontal cortex (dmPFC)
Zuckerman Institute Postdoctoral Seminar (ZIPS, Columbia University, USA

09/2023 **Poster |** Exposure to different mice elicits different representations in the prefrontal cortex. Rajyashree Sen, Alexander Medoff, Mira Vanchiswar, Larry Abbott, and Richard Axel
Howard Hughes Medical Institute Science Meeting, USA

11/2020 **Invited Talk |** Distributed control of motor circuits for backward walking in *Drosophila*
Undergraduate Journal Club in Bucknell University, USA (Virtual)

08/2019 **Invited Talk |** Neuronal control of backward walking in *Drosophila melanogaster*
Indian Institute of Science Education and Research (IISER) Pune, India

05/2018 **Nominated Talk |** Neuronal control of backward walking in *Drosophila*
Junior Scientist Seminar Series, Janelia Research Campus, USA

04/2018 **Short Talk & Poster |** Neuronal control of backward walking in *Drosophila*
Rajyashree Sen, Till Bockemühl, Kaiyu Wang, Michael Dübbert, Salil Bidaye, Ansgar Büschges and Barry Dickson.
Neural Circuits of the Insect Ventral Nerve Cord, Janelia Research Campus, USA

10/2013 **Poster |** Neural control of directed walking in *Drosophila.*
Rajyashree Sen, Salil Bidaye and Barry J. Dickson.
Emerging Techniques to Map and Functionally Characterize Neural Circuits in Drosophila, Janelia Research Campus, USA

10/2013 **Poster |** Neural control of directed walking in *Drosophila melanogaster.*
Rajyashree Sen, Salil Bidaye and Barry J. Dickson.
Neurobiology of Drosophila Meeting, Cold Spring Harbor Laboratory, USA

TEACHING

Lecturer

Fall 2024 — **The Parental Brain (Columbia University's Prison Education Project)**
Adapted and taught a semester-long Columbia University course at **Taconic Correctional Facility** (medium-security women's prison).

Guest Lecturer

Fall 2023, Fall 2024 — **The Parental Brain (Seminar)**
Department of Psychology, Columbia University, USA
Developed a module for the course syllabus and led discussions of scientific papers using active learning techniques.

Training

07/2025 — **Introduction to Writing to Learn in STEM Disciplines**
Institute of Writing and Thinking, Bard College
Completed a weeklong pedagogical training workshop on using writing as a tool for learning in STEM courses.

03/2025 — **Teaching for Transfer: Helping Students Remember What They Already Know**
Institute of Writing and Thinking, Bard College
Completed a one-day pedagogical training workshop on helping students transfer prior knowledge to new contexts.

2017 — **Scientists Teaching Science (Nine Week Course)**
National Institute of Health
Completed a comprehensive pedagogical training course designed to strengthen scientists' teaching skills.

MENTORSHIP

07/2025 – present — **Research Assistant:** Baihe (Lily) Zhang

02/2025 – present — **Undergraduate:** Molly Rose Graff (Barnard College)

02/2025 – 07/2025 — **Undergraduate:** Sarafina Belafonte (Columbia University)

01/2023 – 08/2025 — **Research Assistant:** Alexander Medoff
Current: Graduate student at Columbia University

04/2024 – 11/2024 — **Mentor for Black Undergraduate Mentorship Program (BUMP)**

05/2023 – 06/2023 — **Undergraduate:** Fatma Albwardy (Columbia University)

01/2020 – 05/2023 — **Research Assistant:** Mira Vanchiswar
Current: Graduate student at Cornell University

01/2021 – 05/2023 — **Undergraduate:** Emily Sun (Columbia University)
Current: M.D. candidate, University of Chicago

01/2023 – 04/2023 — **Undergraduate:** Lina Huang (Columbia University)

06/2022 – 08/2023, 01/2023 – 02/2023 — **Undergraduate:** Delaney Bessel (Franklin & Marshall College)
Current: MS Genetic Counseling Student at Long Island University

06/2021 – 07/2021 — **Mentor for Summer Program for Undergraduate Rising Stars (SPURS)**
Supervised Emmanuel Woredekal (Rutgers University)
Current: M.D. candidate, Sidney Kimmel Medical College

01/2018 – 03/2019 — **Women/Non-Binary Peer Mentoring Group**
Janelia Research Campus, USA

TRAINING

2024 — **Research Mentor Training Workshop**
Center for the Improvement of Mentored Experiences in Research (CIMER)
Completed a two-day mentorship workshop designed to enhance the mentoring skills of scientists.

WORKSHOPS

07/2025 – 08/2025 — **Methods in Computational Neuroscience (selected)**
Marine Biology Laboratory, Woods Hole, USA

10/2019 — **Janelia Junior Scientist Workshop (selected)**
Mechanistic Cognitive Neuroscience
Janelia Research Campus, USA

10/2012 – 09/2015 — **Marie Curie FLiACT Training Network Workshops on Fly Neuroscience**
Max Planck Institute for Chemical Ecology, Jena; Max Planck Institute of Neurobiology, Martinsried; KUL, Leuven, Belgium; Center for Genomic Regulation, Barcelona

ACADEMIC SERVICE

01/2025 – present — **Columbia Neuroscience Seminar Series – Systems Neuroscience Selection Committee**
Columbia University, USA

2019 – 2025 — **Invited Reviewer:** eLife, Frontiers in Neuroscience

COMMUNICATION & OUTREACH

IMPROV COMEDY

11/2023 – present — **Producer and performer, *The Lab*:** a science-inspired improv comedy show.

2012 – present — **Entertained large audiences** for over 10 years as a performer with various improv comedy teams, including house teams at Magnet Theater (New York), Washington Improv Theater (DC), and Laugh Index Theater (DC).

07/2024 – 07/2025 — **Second City Conservatory,** *Second City New York, USA*

WRITING

07/2023 — **Writing Merit Scholarship – New York State Summer Writers Institute**
Skidmore College, New York, USA
Nominated by Zuckerman Institute's 2022 writer-in-residence Claire Messud.

PUBLIC ENGAGEMENT

2019 – present — **Invited Guest Scientist (Science & Public Engagement, New York, NY)**
Featured in a range of NYC science communication events, including *Story Collider* podcast, the National Jazz Museum's *Science and Jazz Talk Show*, and miscellaneous science-comedy variety shows.

03/2020 — **Finalist, 2020 iBiology Young Scientist Seminars Series Competition**
iBiology & Alan Alda Center for Communicating Science

---------- REFERENCES ----------

Richard Axel, M.D.
(Postdoctoral Advisor)
Nobel Laureate, 2004
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